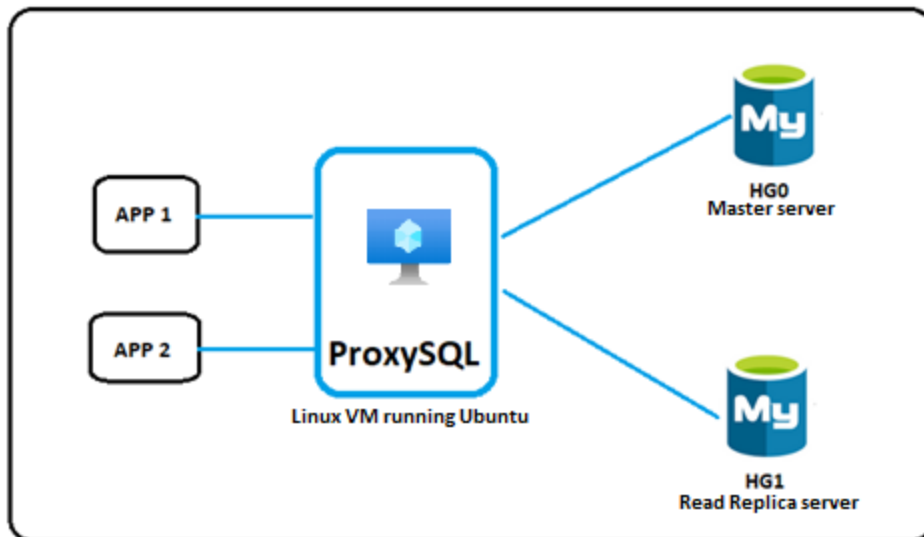


Cấu hình proxySQL

Usecase:



- ProxySQL : 10.4.1.147
- Master: 10.4.1.174
- Slave: 10.4.1.215

Cấu hình Proxy

Lưu ý: Cài đặt dựa trên hướng dẫn của proxySQL

Cấu hình systemd

- Auto restart

```
#/lib/systemd/system/proxysql.service
[Service]
Restart=always
StartLimitIntervalSec=60
StartLimitBurst=3
RestartSec=5
```

- Giới hạn ram

```
#/lib/systemd/system/proxysql.service
[Service]
LimitMEMLOCK=512M
```

- LimitOpenFile

```
#/lib/systemd/system/proxysql.service
[Service]
LimitNOFILE=100000
```

```
#/etc/security/limits.conf
*          soft    nofile    100000
*          hard    nofile    100000
```

```
systemctl daemon-reexec
systemctl daemon-reload
systemctl restart proxysql
```

Truy cập mysql shell quản trị dựa trên ProxySQL guide

- Khai báo server

```
INSERT INTO mysql_servers (hostgroup_id, hostname, port, status)
VALUES
    (1, '10.4.1.174', 3306), -- master
    (2, '10.4.1.215', 3306); -- slave

LOAD MYSQL SERVERS TO RUNTIME;
SAVE MYSQL SERVERS TO DISK;
```

```
LOAD MYSQL SERVERS TO RUNTIME;  
SAVE MYSQL SERVERS TO DISK;
```

- Khai báo rule

```
INSERT INTO mysql_query_rules (rule_id, match_pattern, destination_hostgroup,
apply, active)
VALUES
    (1, '^SELECT', 20, 1, 1),
    (2, '^INSERT|^UPDATE|^DELETE|^REPLACE|^CREATE|^ALTER|^DROP', 10, 1, 1);

LOAD MYSQL QUERY RULES TO RUNTIME;
SAVE MYSQL QUERY RULES TO DISK;
```

```
Admin> SELECT rule_id, match_pattern, negate_match_pattern, re_modifiers, destination_hostgroup, apply, active FROM runtime_mysql_query_rules;
```

rule_id	match_pattern	negate_match_pattern	re_modifiers	destination_hostgroup	apply	active
1	^SELECT	0	CASELESS	20	1	1
2	^INSERT ^UPDATE ^DELETE ^REPLACE	0	CASELESS	10	1	1
3	^CREATE	0	CASELESS	10	1	1
4	^ALTER	0	CASELESS	10	1	1
5	^DROP	0	CASELESS	10	1	1

```
5 rows in set (0.01 sec)
```

- Khai báo user
 - Proxy yêu cầu khai báo các user dùng để kết nối từ client => vDBS cần làm chức năng sync user từ database sang proxy
 - Password có thể lưu dạng plaintext hoặc hash (copy hash từ database hoặc dùng func `MYSQL_NATIVE_PASSWORD()`)

```
INSERT INTO mysql_users (username, password, default_hostgroup, active)
VALUES
    ('user', 'password', 10, 1); -- hostgroup_id 10: mặc định là master

LOAD MYSQL USERS TO RUNTIME;
SAVE MYSQL USERS TO DISK;
```

```
Admin> select * from mysql_users;
```

username	password	active	use_ssl	default_hostgroup	default_schema	schema_locked	transaction_persistent	fast_forward	backend	frontend	max_connections	attributes	comment
maxscale	VtdcQ2824	1	0	10	NULL	0	1	0	1	1	10000		
idcadmin	*D32C36CDB1A412155C7CB1C5BF1F3A08EEED02D5	1	0	10	NULL	0	1	0	1	1	10000		
dbadmin	*22ABA95C4F8FA3E2AA3F361FB17FD725EB1A8BCA	1	0	10	NULL	0	1	0	1	1	10000		

```
3 rows in set (0.00 sec)
```

Test kết nối

Kết nối từ client tới proxy, dùng user của database (đã khai báo ở proxy)

Thực hiện query

Kiểm tra kết quả

```
SELECT * FROM stats_mysql_connection_pool;
```

```
Admin> SELECT * FROM stats_mysql_connection_pool;
```

hostgroup	srv_host	srv_port	status	ConnUsed	ConnFree	ConnOK	ConnERR	MaxConnUsed	Queries	Queries_GTID_sync	Bytes_data_sent	Bytes_data_rcv	Latency_us
10	10.4.1.174	3306	ONLINE	0	0	24	0	2	556	0	113218	437919	1270
20	10.4.1.215	3306	ONLINE	0	0	22	110	2	167	0	21265	78341	1357

2 rows in set (0.00 sec)